End-to-end Handwritten Paragraph Text Recognition Using a Vertical Attention Network [1]

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Introduction

Task: paragraph recognition.

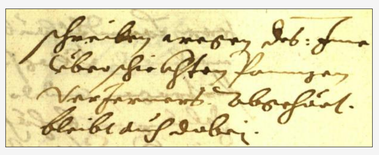
- Traditional approach in 2 steps:
  - Text line segmentation / text line recognition.
- Our unified end-to-end approach:
  - Recurrent line-by-line recognition process.
  - State-of-the-art results on RIMES, IAM and READ 2016 datasets.

Our code and trained model weights are publicly available:
`https://github.com/FactoDeepLearning/VerticalAttentionOCR`

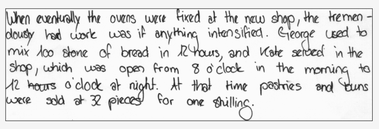
Datasets: paragraph handwritings

Dataset	Training	Validation	Test	Charset size
RIMES	1,400	100	100	100
IAM	747	116	336	79
READ 2016	1,584	179	197	89

- Images are downscaled to 150 dpi.
- The FCN Encoder is pretrained with line-level images.

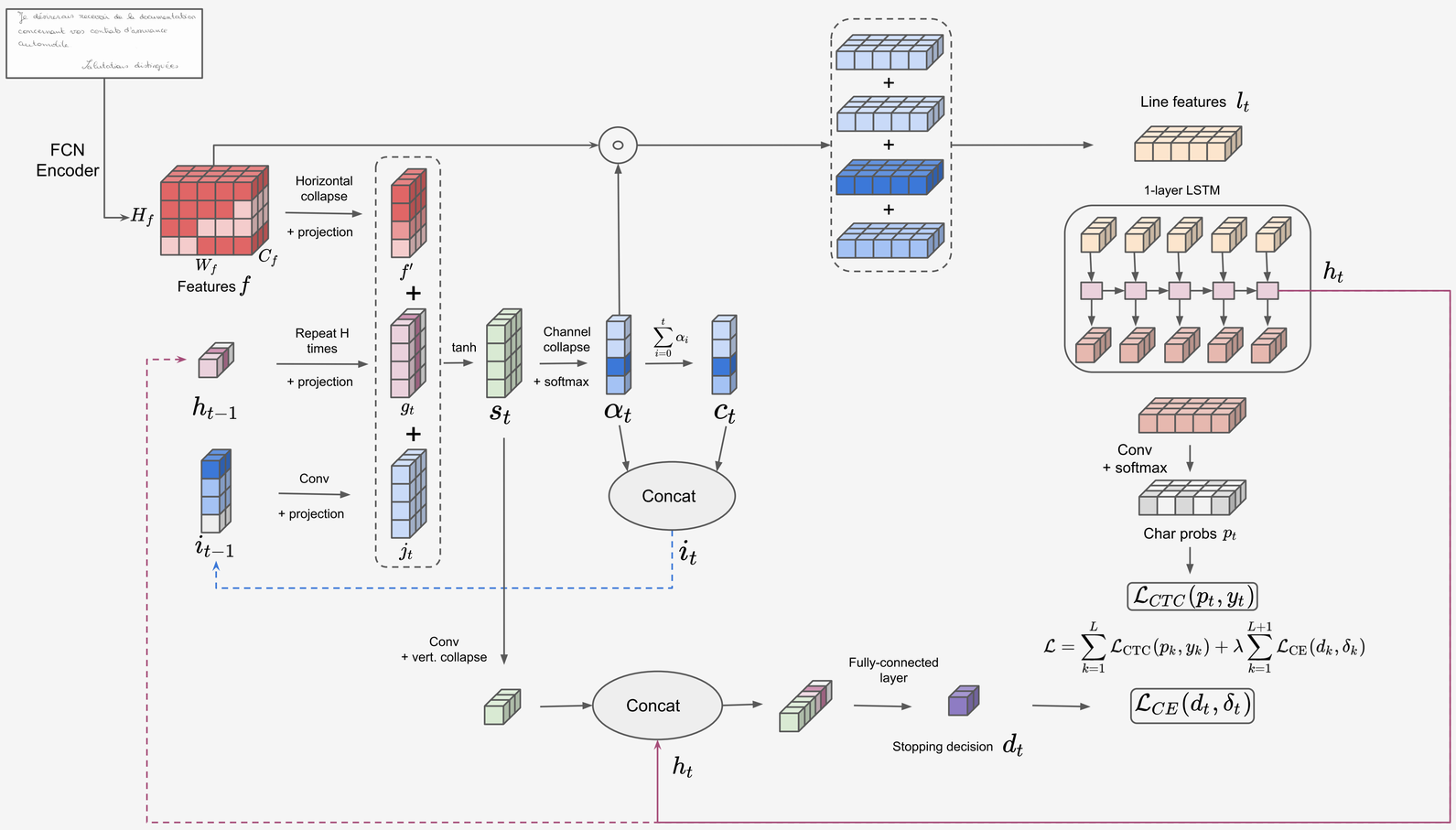


READ 2016 example.

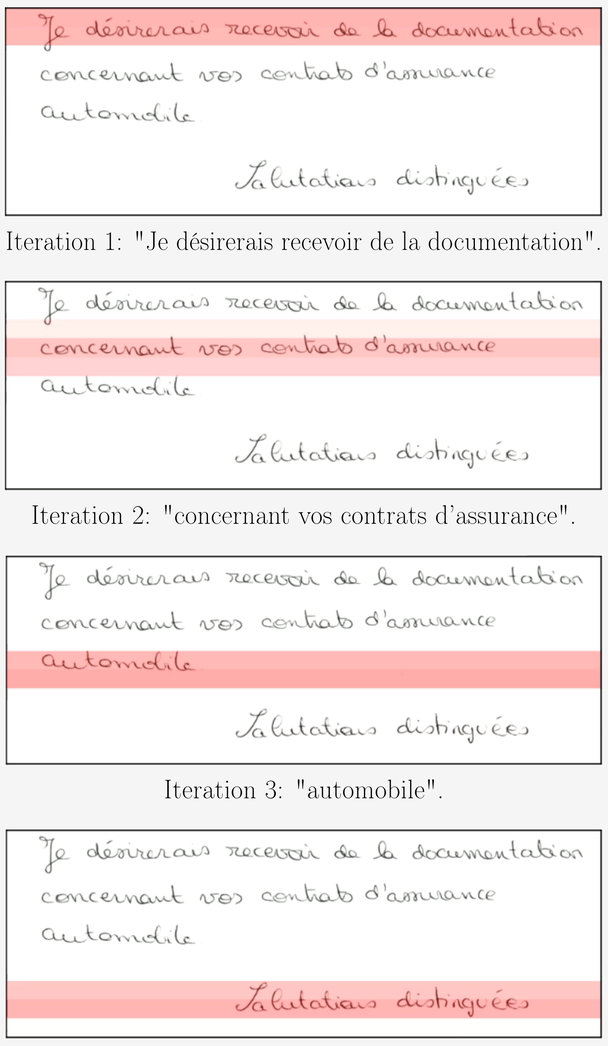


IAM example.

Architecture: encoder-decoder with hybrid attention



(a) Architecture overview.



Iteration 1: "Je désirerais recevoir de la documentation".

Iteration 2: "concernant vos contrats d'assurance".

Iteration 3: "automobile".

Iteration 4: "Salutations distinguées".

(b) Visualization with a RIMES example.

- Step 1: An FCN encoder extracts 2D features f from the paragraph image.
- Step 2: Vertical attention weights α_t are computed, in a hybrid way, to select the features to focus on.
- Step 3: Line features l_t are computed as a weighted sum between the attention weights α_t and the features f.
- Step 4: End-of-paragraph detection probabilities d_t are computed. Step 4' is carried out if this detection is negative.
- Step 4': Character and CTC blank label probabilities are computed for each frame of the line features l_t through LSTM and convolution.

Step 2 to 4' are repeated until the detection of the end of the paragraph.

Results

Architecture	RIMES		IAM		READ 2016	
	CER (%)	WER (%)	CER (%)	WER (%)	CER (%)	WER (%)
[2] CNN+MDLSTM*	2.9	12.6	7.9	24.6		
[3] CNN+MDLSTM**			16.2			
[4] RPN+CNN+BLSTM	2.1	9.3	6.4	23.2		
[5] GFCN			4.7			
This work [1] FCN+LSTM*	**1.90**	**8.83**	**4.32**	**16.24**	**3.63**	**16.75**

* with line-level attention ** with character-level attention

Our approach:

- Best path decoding.
- No language model.
- No lexicon constraints.
- Same hyperparameters for all datasets.

Conclusion

- A new approach for handwritten text recognition at paragraph level: end-to-end & segmentation-free.
- The model is trained with a hybrid loss:
  - The CTC loss for text recognition.
  - The Cross-Entropy loss for end-of-paragraph detection.
- State-of-the-art results on 3 datasets: IAM, RIMES and READ 2016.

Paper, code and more !

References

[1] Denis Coquenet, Clement Chatelain, and Thierry Paquet. "End-to-end Handwritten Paragraph Text Recognition Using a Vertical Attention Network". In: *IEEE TPAMI* (2022).
[2] Théodore Bluche. "Joint Line Segmentation and Transcription for End-to-End Handwritten Paragraph Recognition". In: *NIPS*. 2016, pp. 838–846.
[3] Théodore Bluche, Jérôme Louradour, and Ronaldo O. Messina. "Scan, Attend and Read: End-to-End Handwritten Paragraph Recognition with MDLSTM Attention". In: *ICDAR*. 2017, pp. 1050–1055.
[4] Curtis Wigington et al. "Start, Follow, Read: End-to-End Full-Page Handwriting Recognition". In: *ECCV*. Vol. 11210. Lecture Notes in Computer Science. 2018, pp. 372–388.
[5] Mohamed Yousef and Tom E. Bishop. "OrigamiNet: Weakly-Supervised, Segmentation-Free, One-Step, Full Page Text Recognition by learning to unfold". In: *CVPR*. 2020, pp. 14698–14707.